Cape Peninsula University of Technology

Faculty of Engineering and the Built Environment
Department of Electrical, Electronic and Computer Engineering

Software Design 1 (SDN150S)
Project Brief
Due Date – November 08, 2026

INSTRUCTIONS:

- Write a technical report of not more than 2000 words to management, discussing your approach to the implementation of the assigned mini project. (Font: Cabri 11, Line space: 1.5, Citation: Harvard style)
- The report should have an introduction, sub-questions, discussions, result/analysis, and conclusion section. Discuss your chosen software design approach, engineering method, skills, and tools used for implementation.
- The report should contain a pseudocode, a UML class or activity diagram, and java program.
- Kindly note that your java program must be implemented using the following established concepts: Constructor, Modifiers, Control statements, and Java Frame Forms.
- Remember to use white spaces and single/multiline comments to ensure readability.
- Marks will be based on technical report, code implementation, and group participation.

Project Overview

This project aims to develop your skills in creating practical, user-friendly applications using Java. You'll gain hands-on experience in software design, implementation, and documentation. This project will enhance your problem-solving abilities, teamwork skills, and understanding of the software development lifecycle. Your team will implement a java application for one of the assigned project areas listed below. Each application is designed to solve a specific problem or provide utility in a particular domain. Your task is to create a functional, user-friendly application that meets the given requirements and demonstrates your understanding of key Java programming concepts.

Requirements

1. **Application Development:**
    - Develop your chosen application using Java on an appropriate IDE.
    - Implement a user-friendly graphical interface using Java Frame Forms.
    - Ensure your application includes necessary operations as per the project description.
2. **Code Implementation:**
    - Use established Java concepts: Constructors, Modifiers, Control statements, and Java Frame Forms.
    - Implement error handling and input validation where necessary.
    - Use white spaces and single/multiline comments to ensure readability.
3. **Technical Documentation:**
    - Provide a technical report (max 2000 words) including: introduction, sub-questions, discussions (including chosen software design approach, engineering method, skills, and tools used), results/analysis, conclusion.
    - Include a pseudocode for your application.
    - Create a UML class or activity diagram illustrating your application's structure or flow.
4. **Version Control:**
    - Use Git or any repository for version control throughout your development process.
    - Maintain the repository for your project with regular, meaningful commits/updates.

Submission Requirements

1. Source code (Java Program)
2. Technical report (PDF format, Font: Cabri 11, Line space: 1.5, Citation: Harvard style)
3. UML diagram (as part of the report)
4. Pseudocode (as part of the report)
5. Video demonstration (3-5 minutes)

Group Projects List

1. Consider a scenario of a where lecturers, admin staff, and authorities need to go through the records of students for various purposes like searching for a particular student's details. Manually going through records is a tedious job and time-consuming. Hence, it is better to have a software that allows users to insert, update, search, or delete records automatically. Hence, your team have been tasked with designing a Java program of a student record management system for CPUT faculty of engineering and the built environment. The design should enable the creation of new student records, store the personal and academic records of each student, allow all student records to be accessed by authorizes, but restrict access to lectures and admins (e.g., registration, courses enrollment).

2. An interactive Automated Teller Machines (ATMs) that can disburse money and take deposits. It aids in financial literacy and makes formal financial services more accessible to people in rural areas. As computer engineers, your team have been tasked with design a Java program that illustrates the operation of an ATM. The program should allow an authenticated user to choose from the options displayed on the ATM screen. The options should be: Withdraw, Deposit, Check Balance, and Exit.
    - Assume the user has an initial account balance of R10 000.00.
    - To withdraw money, obtain the user's withdrawal amount, deduct it from the total balance, and print the successful message.
    - To deposit money, ask the user for the deposit amount, add it to the total balance, and print the successful message.
    - To check the balance, publish the user's total balance.
    - To exit the current Transaction mode and return the user to the home page or initial screen, use the System.exit(0) method.

3. Efficient communication is vital in many settings, from offices to classrooms. As computer engineers, you team is tasked with developing a Java program for a Simple Chat Application that operates over a local network. The application should support real-time messaging between multiple users, private messaging, and basic file sharing. Consider implementing features like chat rooms, message encryption, and the ability to save chat logs.

4. The Currency Converter is a piece of software that allows you to convert currency values quickly and easily using current exchange rates. These currency converters are commonly accessed for free on the Internet, and they can swiftly convert the value of one currency to another, such as dollars to pounds, rand to dollar, and back. Your team have been tasked with designing a Java program that converts 10 cryptocurrency pairs (e.g., BTC/ETH, BTC/LTC, etc.).

5. Text editors are fundamental tools for writers, programmers, and anyone who works with text documents. As computer engineers, your team are tasked with creating a Java-based text editor that provides essential functionalities such as creating, opening, editing, and saving text files. Your editor should include features like find and replace, word count, and basic text formatting. Consider adding more advanced features like syntax highlighting for programming languages or a spell-check function to enhance its utility.

6. Scientific programable tools are a nifty way to perform certain complex computations with little effort. Your team have been tasked with designing a Java program of a matrix calculator, which will be used by first year students in the Bachelor of Engineering Technology (BET) stream at CPUT.

7. As our lives become increasingly busy, effective task management is crucial. Your team have been tasked with designing a Java program for a To-Do List application. This software should allow users to add, delete, and prioritize tasks, set due dates, and mark items as complete. The application should help users organize their personal and professional responsibilities, improving productivity and reducing stress. Consider implementing features such as task categories, reminders, and the ability to sync across devices.

8. Flight reservation system helps an airline firm increase sales and gain a competitive advantage. It allows clients to make reservations at their leisure regardless of whether it is within business hours or not. As computer engineer, your team have been tasked with designing a Java program of a flight reservation system to be used by FlySafair airlines. The system should require:
    - The user to pick the flight type, such as international or domestic, before entering the reservation details, such as source and destination stations.
    - Tell the user if seats are available for that flight.
    - Notify the user and provide a printout of the reservation ticket.
    - Store reservation details and retrieve information using a database.

9. In the digital age, quick note-taking is an essential part of information management. Your team task is to develop a Java-based Notepad application that allows users to create, edit, and organize digital notes. The application should support multiple note creation, categorization, and search functionality. Consider implementing features like rich text formatting, image insertion, and cloud synchronization to make the notepad more versatile and user-friendly.

10. Libraries undertake a variety of duties, including collecting books, organizing them in a logical sequence, and preserving them. The most crucial component of library management is making books available to readers. As computer engineer, your team have been tasked with designing a Java program for a library management system to controls all aspects of CPUT's library's operations. Your design should assist librarians in maintaining a database of new books as well as books borrowed by students and their due dates. All library activities should be automated using this system.

11. In today's connected world, managing contact information efficiently is crucial. Your team is tasked with designing a Java program for an Address Book application. This software should allow users to store, organize, and quickly retrieve contact information including names, phone numbers, email addresses, and physical addresses. Consider implementing features such as contact grouping, search functionality, and the ability to import/export contacts in various formats (e.g., vCard, txt, CSV).

12. Time management is a critical skill in both personal and professional life. Your team task is to create a Java-based Appointment Scheduler that helps users manage their time effectively. The application should allow users to create, edit, and delete appointments, set reminders, and view their schedule in different formats (daily, weekly, monthly). Consider implementing features like recurring appointments, conflict detection, and integration with external calendars.

13. Electronic voting technology aims to speed up ballot counting, minimize the expense of hiring employees to manually tally ballots, and increase accessibility for disabled voters. Expenses are also predicted to reduce in the long run. Results can be reported and published in a shorter amount of time. As computer engineers, your team have been tasked with designing a Java program of an automated voting system, which is aimed to speed up ballot counting in the next student representative council (SRC) election in CPUT.

14. Efficient grade management is crucial for educators to monitor student progress and provide timely feedback. As computer engineers, your team have been tasked with designing a Java program for a Student Grade Tracker. This application should allow teachers to input and manage student grades across multiple subjects, calculate GPAs, generate performance reports, and track attendance. Consider implementing features like customizable grading scales, weighted grade calculations, and the ability to export reports in various formats.

15. Interactive learning tools can significantly enhance the educational experience. Your team task is to develop a Java-based Quiz Application that allows educators to create and administer quizzes, and students to take them. The application should support various question types (multiple choice, true/false, short answer), implement a scoring system, and provide immediate feedback. Consider adding features like timed quizzes, randomized question order, and the ability to track student performance over time.

16. Accurate time measurement is crucial in many scenarios, from sports to cooking. Your team task is to create a Java-based Stopwatch and Timer application. The program should include a stopwatch function with lap timing capabilities, as well as a countdown timer with customizable alarms. Consider implementing features like multiple concurrent timers, a Pomodoro timer for productivity, and the ability to save and name different timer presets.

17. Mathematics is a fundamental subject that often benefits from interactive learning methods. Your team have been tasked with creating a Java program for a Math Quiz Game aimed at high school students. The game should generate questions covering various mathematical concepts, track scores, and adjust difficulty based on the user's performance. Consider implementing different game modes (e.g., timed challenges, practice mode), visual aids for complex problems, and a reward system to encourage continued learning.

18. Calculators are essential tools in many fields, from basic arithmetic to complex scientific calculations. Your team task is to design a Java program for a versatile calculator. The application should perform basic arithmetic operations, as well as more advanced functions like trigonometry, logarithms, and statistical calculations. Consider implementing features such as a history log, customizable unit conversions, and the ability to save and recall frequently used calculations.

19. Access to accurate and timely weather information is important for daily planning and safety. Your team have been tasked with developing a Java program for a Weather Application. The app should provide current weather conditions and short-term forecasts for user-specified locations. Utilize a weather API to fetch real-time data. Consider implementing features such as severe weather alerts, historical weather data, and the ability to save multiple locations for quick access.

20. Digital art tools have revolutionized the way we create and edit images. Your team task is to design a Java-based Paint Program that allows users to create digital artwork. The application should include basic drawing tools (brush, eraser, shapes), color selection, and the ability to save and load images. Consider implementing more advanced features like layers, filters, and basic image editing tools (crop, resize, rotate).

21. Efficient image management and viewing is important for both casual users and professionals. Your team task is to develop a Java-based Image Viewer application. The program should support viewing common image formats, basic image manipulation (rotate, flip, zoom), and simple file management. Consider implementing features like slideshows, basic editing tools (crop, adjust brightness/contrast), and the ability to add tags or comments to images.

22. Effective inventory management is crucial for businesses of all sizes. Your team have been tasked with designing a Java program for a Simple Inventory System. This application should allow users to add, update, and remove items from the inventory, track stock levels, and generate reports. Consider implementing features like low stock alerts, barcode scanning integration, and the ability to categorize items and track their location within a warehouse.

23. File management is a fundamental task in computing. Your team task is to create a Java-based File Explorer application. The program should allow users to navigate their file system, perform basic file operations (copy, move, delete, rename), and provide a search function. Consider implementing features like file previews, compression/decompression capabilities, and the ability to connect to cloud storage services.

24. Automated customer service can significantly enhance the online shopping experience. Your team task is to create a Java-based Chat Bot for an e-commerce platform. The bot should be able to answer frequently asked questions, assist with product searches, and provide basic order tracking information. Consider implementing natural language processing capabilities, the ability to escalate to human customer service when needed, and a learning mechanism to improve responses over time.

25. Simple games can be excellent projects for learning programming concepts. Your team have been tasked with designing a Java program for a Tic-Tac-Toe game. The game should feature a graphical user interface, support both two-player and AI opponent modes, and keep track of wins and losses. Consider implementing multiple difficulty levels for the AI, a larger grid size option, and the ability to customize game tokens.

Marking Rubric

CODE IMPLEMENTATION (40 %)

S/n	Items	Deliverables	Score
1	**Pseudocode** (An informal description of a program that is void of any strict programming syntax)	Presentation: Clear and descriptive. No ambiguity and without use of program syntax.	**5**
		Correctness: A simple grammatical representation of the implemented code	**10**
2	**UML diagram** (A graphical representation of sequence and steps to solving a problem)	Presentation: neatness and clarity of activity or class diagram	**5**
		Correctness: provides accurate graphical representation of the implemented code	**10**
3	**Code Implementation** (Accurate use of programming syntax and logic to implement solution)	Implementation to utilize specified concepts such as constructors, modifiers, controls structures, and Java Frame forms.	Constructors/modifiers = 5 control structures = 5 JFrame Form = 10 Total: **20**
		Documentation: provides author information, makes variable descriptions are clear and meaningful with comments	**5**
		Correctness: accurate use of programmable syntax in-line with algorithm and/or UML diagram	**10**
		Presentation: Single, multiline comments and white spaces are used to ensure neatness and readability of code.	**10**
		Viability: successful testing of implemented solution	**10**
4	**Demonstration** (A brief video demonstration (3-5 minutes) showcasing your application's features)	Functionality: demonstrate the practicality of your implemented solution	**15**
		Final Mark (%)	**100**

TECHNICAL REPORT (40 %)

GA6: Professional and technical communication:
Software Design 1
SDN150S

	Surname:		**Initial:**		**Student No:**	
	1 (0-25%)	**2** (26-49%)	**3** (50-75%)	**4** (76-100%)	**Assessor's Rating**	**Student Score (%)**
INDICATORS & WEIGHTING	**Needs work**	**Developing**	**Competent**	**Strong**		
1. Purpose: Ability to identify the reason for the communication, choose the best communication for the task, and use the purpose to guide the communication. (10%)	Purpose is unclear. Does not guide the communication. Choice of communication is ineffective.	Purpose is somewhat clear and somewhat guides the communication. Choice of communication is somewhat effective.	Purpose is evident, and mostly guides the communication. Choice of communication is effective.	Purpose is clear and effectively guides the communication. Choice of communication is optimal.	4	**10**
2. Main Idea: Ability to communicate the main idea. (10%)	Main idea is difficult to understand.	Main idea is somewhat understandable.	Main idea is understandable.	Main idea is clearly and precisely stated.	4	**10**
3. Supporting Materials: Ability to use information sources (texts, journals, research) and/or generate materials (examples, statistics, analogies) to support the purpose and main idea, and establish credibiity. (0%)	Demonstrates minimal or no attempt to use trustworthy, appropriate and pertinent materials to support the purpose and/or main idea.	Demonstrates an attempt to use trustworthy, appropriate and pertinent materials that support and develop the purpose and main idea.	Demonstrates consistent use of trustworthy, appropriate and pertinent materials that support the purpose and main idea.	Demonstrates skillful use of superior, trustworthy, pertinent and compelling materials that support the purpose and main idea.	4	**10**
4. Organization: Ability to structure ideas to move logically forward. E.g., For written communication both macro (introduction, paragraphs, sequence of content, conclusion) and micro organization (sentences, transitions) are considered. (10%)	Material has little logical order, and is often unclear, inconherent and difficult to follow.	Materical has some logical order and is somewhat coherent or easy to follow.	Material moves logically forward, and both macro and micro elements are clear, coherent, and easy to follow.	Materials are seamlessly arranged in a logical sequence and both macro and micro elements enhance the work and the reader's comprehension.	4	**10**
5. Tailoring Communication: Ability to use language that is appropriate for: (i) the genre (written, verbal or electronic communications, i.e., texts & emails); (ii) the target audience; (iii) the company/persons that the communicator is representing. (10%)	Minimal or no consideration of the genre, the target audience or the company/persons represented is taken within the communication. Communication contains details or technical content that are unsuitable.	Some consideration of the genre, the target audience and the company/persons represented are taken within the communication. Communication may contain some detail or technical content that are unsuitable.	Communication is suitable for the genre, the target audience and the company/persons represented.	Communication is skillfully crafted to suit the genre, the level of target audience, and the company/persons represented.	4	**10**
6. Clarity and Conciseness: Ability to use language that clearly and concisely conveys meaning and supports the purpose of the work. (10%)	Uses language that may be unclear and/or inconcise, may sometimes impede meaning, and minimally supports the purpose of the work.	Uses language that generally conveys meaning, is somewhat concise, and supports the purpose of the work.	Uses language that clearly and concisely conveys meaning, and supports the purpose of the work.	Uses language that conveys meaning with clarity and parsimony, and enhances the purpose of the work.	4	**10**
7. Mechanics: Ability to use language that is mechanically correct (punctuation, spelling, grammar). (10%)	Communication contains many errors. Errors impede meaning.	Communication contains errors. Errors may be distracting.	Communication contains some errors, but errors do not detract from meaning.	Communication contains very few or no errors.	4	**10**

8. Genre and Conventions: Ability to understand and use conventions intrinsic to engineering genres (formal/informal reports, lab reports, final completion reports, proposals, presentations). (10%)	Demonstrates minimal or no understanding of or effort to use the conventions inherent within the engineering genre.	Demonstrates some understanding of and attempts to use the conventions inherent within the engineering genre.	Demonstrates understanding of and use of the conventions inherent within the engineering genre.	Demonstrates thorough/sophisticated understanding of and skillful use of the conventions inherent within the engineering genre.	4	**10**
9. Illustrations (Graphs, Tables, Figures and Diagrams): Ability to use illustrations properly to support ideas (citations, position on page, integration, design and support of ideas). (10%)	Illustrations are used, but minimally support ideas, and are not properly cited, positioned on page, integrated and/or designed. They do not support, explain or interpret information.	Illustrations are for the most part properly used to support ideas (an attempt is made to correctly cite, position on page, integrate, and/or design, and to support, explain or interpret information).	Illustrations are properly used to support ideas (correctly cited, positioned on page, integrated and designed; they support, explain and interpret information).	Illustrations are skillfully used to support ideas (correctly cited, skillfully positioned on page, well integrated and designed; they enhance points, explain, interpret, and assess information).	4	**10**
10. Formatting/Layout/Design of Communication: Ability to use appropriate or precribed format, which is effectively designed, clearly labeled, neatly and professionally presented. (10%)	The appropriate/ prescribed format is not followed. Design of communication is ineffective; headings are unclear, and work is not neatly presented. Format does not invite reading.	The appropriate/ prescribed format is followed in most portions of the communication. Design is somewhat effective; headings are somewhat clear, and work is for the most part, neatly presented.	The appropriate/ prescribedformat is followed. Design is effective; headings are clear, and work is neatly and professionally presented. Format invites reading.	The appropriate/prescribed format is precisely followed. Design is exemplary. Headings are clear and add to the overall meaning of, and/or enhance the communication. Work is exceptionally neat and professionally presented. Format compels reading.	4	**10**
				Final Mark (%)	**100**	

Examiner:	MR S.O EKWE	
Internal moderator:		
External moderator:		

GROUP PARTICIPATION (20 %)

GA5: Engineering methods, skills, tools, including Information technology:

Software Design 1

SDN150S

	Surname:		**Initial:**		**Student No:**	
	1 (0-25%)	**2** (26-49%)	**3** (50-75%)	**4** (76-100%)	**Assessor's Rating**	**Student Score (%)**
INDICATORS & WEIGHTING	**Needs work**	**Developing**	**Competent**	**Strong**		
1. Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools. (20%)	Demonstrates minimal or no ability to describe and/or explain the principles behind and applicability of engineering tools.	Demonstrates some ability to describe and/or explain the principles behind and applicability of engineering tools.	Demonstrates ability to describe and explain the principles behind and applicability of engineering tools.	Demonstrates skillful ability to describe and explain the principles behind and applicability of engineering tools.	4	**20**
2. Identify and Use Tools: Ability to identify and use relevant tools for an engineering activity. (20%)	Demonstrates minimal or no ability to identify or use tools for an engineering activity.	Demonstrates some ability to identify and use tools for an engineering activity, but may not identify the most relevant tool.	Demonstrates an ability to identify and use relevant tools for an engineering activity.	Demonstrates skillful ability to identify and use the most relevant tools for a range of engineering activities.	4	**20**
3. Create Tools: Ability to create engineering tools. (30%)	Demonstrates minimal or no ability to create engineering tools.	Demonstrates some ability to create engineering tools with guidance.	Demonstrates an ability to create engineering tools.	Demonstrates skillful ability to create engineering tools.	4	**30**
4. Evaluate Tools: Ability to identify the limitations in the use of engineering tools, and their underlying assumptions. (30%)	Demonstrates minimal or no ability to identify the limitations of tools and understand the assumptions.	Demonstrates some ability to identify the limitations of tools and some understanding of the assumptions.	Demonstrates the ability to identify the limitations of tools and understands the assumptions.	Demonstrates skillful ability to evaluate the limitations of tools and discusses the assumptions.	4	**30**
				Final Mark (%)	**100**	

Examiner:	MR S.O EKWE	
Internal moderator:		
External moderator:		